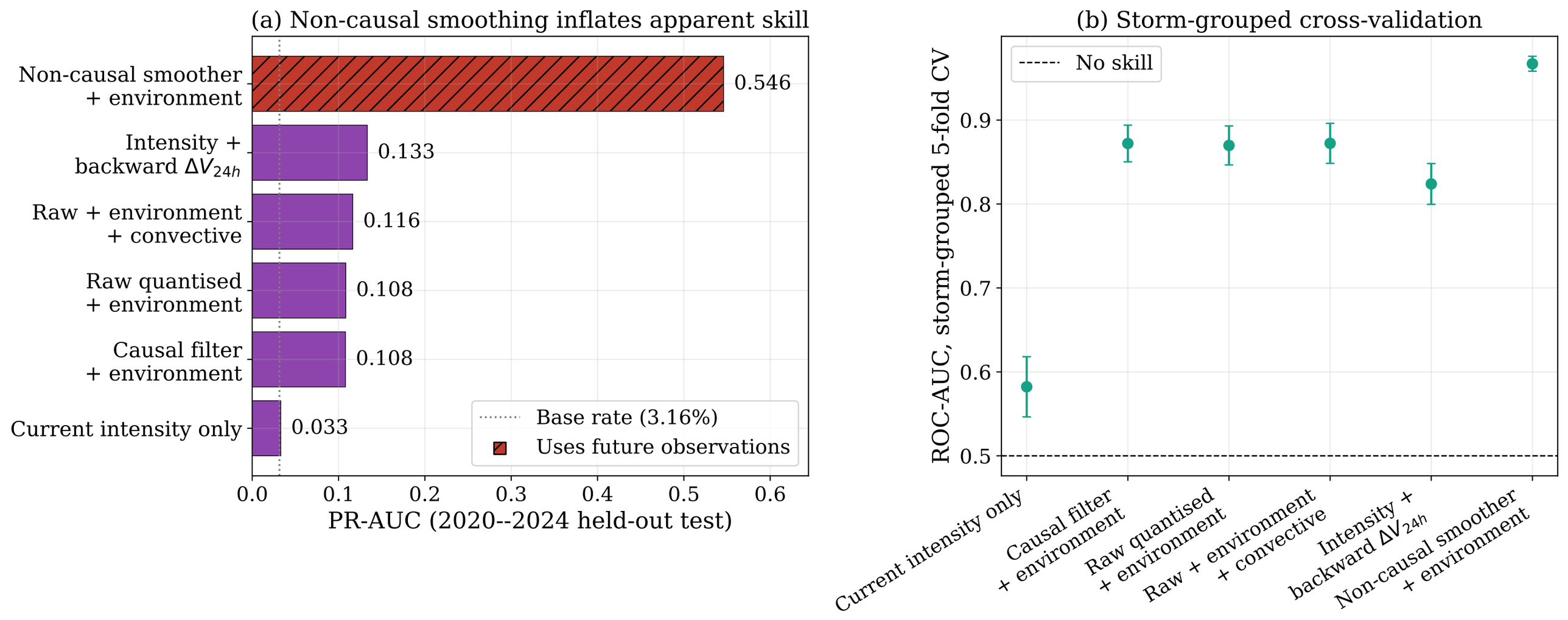
(a) Non-causal smoothing inflates apparent skill
Non-causal smoother + environment
Intensity + backward ΔV_{24h}
Raw + environment + convective
Raw quantised + environment
Causal filter + environment
Current intensity only
0.546
0.133
0.116
0.108
0.108
0.033
Base rate (3.16%)
Uses future observations
0.0
0.1
0.2
0.3
0.4
0.5
0.6
PR-AUC (2020--2024 held-out test)
(b) Storm-grouped cross-validation
No skill
ROC-AUC, storm-grouped 5-fold CV
0.5
0.6
0.7
0.8
0.9
Current intensity only
Causal filter + environment
Raw quantised + environment
Raw + environment + convective
Intensity + backward ΔV_{24h}
Non-causal smoother + environment